



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Spring Semester Examination 2022-23

Date of Examination: 17-02-2023

Session: (AN)

Duration: 2 hrs. Full Marks: 50

Subject No.: CS60075___

Subject: Natural Language Processing

Department/Center/School: CSE

Specific charts, graph paper, log book etc., required : NA

Special Instructions (if any):

[Note: For this question paper, use base 10 for any log computation]

There are 13 questions. All of the questions are compulsory. There are no clarifications. In case of any doubts, please make a valid assumption, write that explicitly and proceed.

1. How would you estimate $P(w_n|w_1, w_2 \dots w_{n-1})$ using a trigram language model? Just write down the expression. [2 marks]
2. Suppose you are using beam search decoding for sequence to sequence with a beam size of 5. At the fourth step in the decoder, i.e., while you are generating the fourth word in the sequence, what is the number of hypotheses for which you will have to compute the log probability? Suppose that the normal length of the generated sentence is much larger as compared to 4. [2 marks]
3. How would you estimate $P(w_1, w_2 \dots w_{n-1}, w_n)$ using a k-gram language model? Just write down the expression. [2 marks]
4. Suppose you are using 3-dimensional word embeddings, and trying to solve analogy task, i.e. find x in "*man*: *woman* :: *prince*: x ". Suppose the embeddings for *man*, *woman* and *prince* are $[0.5 \ -0.3 \ 0.8]$, $[0.3 \ 0.1 \ 0.7]$ and $[0.1 \ -0.5 \ 0.4]$, respectively. What would be the best guess for the embedding of x ? [3 marks]
5. Suppose two words "government" and "debt" co-occur in a corpus of 50,000 documents 100 times, while the word "government" occurs 2,500 times and "debt" occurs 500 times. What will be the pointwise mutual information between these two words? [3 marks]
6. Suppose you are using sequence to sequence models with attention. Your hidden layer is of size 2 (both encoder and decoder). Suppose the encoder has 4 hidden states (2-dimensional) $[1, -1]$, $[-1, 1]$, $[0, -1]$, $[-1, 0]$ and the decoder hidden state at a given time point is $[-1, -1]$. Using attention (dot product) over the encoder states, what will be the concatenated vector to the decoder hidden state? [3 marks]
7. Consider the problem of predicting the next word from a sequence of words using a fixed window neural language model (using the previous 2 words to predict the next word). Suppose, there are overall 100 words in your vocabulary. And each word is represented using a one-hot vector. Suppose the hidden layer has '25' dimensions, what are the total number of parameters that you will have to learn? Ignore the bias terms for this problem. [4 marks]

8. Given a corpus C, the bigram probability of "paper | question" is 0.3 and the count of occurrences of the word "question" is 600. For the same corpus, the likelihood of "paper | question" after applying add-one smoothing is 0.1. What is the vocabulary size of C? [4 marks]
9. Suppose you are using Bi-LSTM for language labeling in code-switched data, and there are 5 possible language tags per word. Assume that each word has a 50-dimension embedding, and the hidden state for both the forward and backward LSTMs are 40-dimension each. How many total parameters will need to be trained? Ignore the bias terms. [5 marks]
10. Suppose you are using hierarchical softmax to compute the probability of the context word given the center word. Suppose, as a given time during training, the center word embeddings be $[0.5 \ 0.3 \ -0.4 \ 0.2]$, and the path to the context word goes via three nodes starting from root (left, right, right) having embeddings as $[0.3 \ 0.1 \ -0.1 \ 0.05]$, $[-0.4 \ 0.5 \ 0.3 \ -0.6]$ and $[0.5 \ 0.3 \ -0.8 \ -0.9]$, respectively. What will be the probability of the context word given the center word? [5 marks]
11. Suppose you are using a GRU for sentence representation, and you are using a 6-dim hidden state. At a particular point in the sequence, you wish to preserve the complete information from the previous state for the 2nd and 4th dimension, while you want to erase the previous information for the 3rd dimension. For the other dimensions, you want the model to behave like an RNN. Write down a possible update and reset vector that will allow you to do this. *[If there are more than one possible answers, you just need to write one of those, do not provide all possible answers.]* [5 marks]
12. Suppose you are computing the word vectors using Skip-gram architecture. You have 5 words in your vocabulary, {passed, through, relu, activation, function} in that order and suppose you have the window, 'through relu activation' in your corpora. You use this window with 'relu' as the center word and one word before and after the center word as your context. Also, suppose that for each word, you have 2-dim in and out vectors, which have the same value at this point given by $[-1, -1], [1, 1], [2, -1], [1, -1], [1, 0]$ for the 5 words, respectively. As per the Skip-gram architecture, what would be the total loss for this window? [6 marks]
13. Suppose you have a vocabulary of size 5 and during decoding, the output vector is $[3, -1, 2, 1, -2]$. Write down the effective probability distribution when you use the following sampling strategies. [6 marks]
 - a. Random sampling with temperature 0.5
 - b. Top-2 sampling
 - c. Nucleus sampling with $p = 0.5$